POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 4001 | Semester IV | Theory | 4 Credits

# Humanities & Social Sciences

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

| | |
|---|---|
| Programmes | **Common to Diploma Programmes** |
| Contact hours | **60** |
| Teaching scheme | **3:1:0:0** |
| Credits | **4** |
| CIE / ESE | **Theory CIE 40 / Theory ESE 60** |
| Self-learning | **5.5 h/week** |

CURRICULUM INTEGRITY

## Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 4001 as Humanities & Social Sciences in Semester IV across multiple diploma programmes. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage is based on public humanities/social-science and engineering-ethics curriculum material; it is not represented as recovered official SITTTR Course 4001 detailed-syllabus wording.

**Source treatment:** The title is common across programmes, so one shared handbook is used. Teaching scheme, credits and assessment values are provisional placeholders. This transparent engineering-context HSS framework must be reconciled with restored official SITTTR details.

**Verified source note:** The official detailed source was inaccessible. Public curriculum descriptions used to frame human experience, communication, ethics, society and environment are listed in References.

COURSE DASHBOARD

## What You Are Expected to Learn

**60**
Contact hours

**4**
Credits

**Theory CIE 40**
CIE

**Theory ESE 60**
ESE

### Course introduction

Humanities & Social Sciences helps engineering students understand people, society, communication, ethics and responsible development. It strengthens evidence-based reasoning, professional conduct and awareness of social and environmental consequences of technical decisions.

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### Prerequisite foundation

No specialist prerequisite; openness to reflection, reading, discussion and evidence-based reasoning.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

**Teaching and assessment scheme**

| L | T | P | S | Self-learning/week | Contact hours | Credits | CIE | ESE |
|---|---|---|---|---|---|---|---|---|
| 3 | 1 | 0 | 0 | 5.5 h/week | 60 | 4 | Theory CIE 40 | Theory ESE 60 |

STUDY METHOD

## How to Use This Handbook

### 1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

### 2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

### 3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

### 4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

# Objectives and Course Outcomes

### Official course objectives

1. Explain humanities and social-science perspectives in engineering practice.
2. Communicate clearly and collaborate responsibly with diverse stakeholders.
3. Apply ethical reasoning to technology and professional decisions.
4. Analyse social and environmental impacts of engineering proposals.

### Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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**Official Course Outcomes**

| CO | Official outcome | Bloom level |
|---|---|---|
| CO1 | Explain how humanities and social sciences inform responsible engineering practice. | Understand |
| CO2 | Apply clear communication and collaborative practices in technical contexts. | Apply |
| CO3 | Analyse ethical dilemmas using stakeholders, consequences, duties and professional responsibility. | Analyse |
| CO4 | Evaluate social and environmental impacts of a technology or engineering proposal. | Evaluate |

| Official CO-PO articulation matrix | | | | | | | |
|---|---|---|---|---|---|---|---|
| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7-PO11 |
| CO1 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO2 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO3 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO4 | 3 | 2 | 2 | 2 | 2 | – | – |

SYLLABUS COVERAGE

# Curriculum Coverage Map

**All official major topics mapped to handbook chapters**

| Module | Title | Official major topics | Hours | CO | Handbook section |
|---|---|---|---|---|---|
| 1 | Humanities, Society and Engineering | Human experience, culture, society, institutions, technology, engineering context and critical inquiry. | Approx. 14 hours | CO1 | Module 1 |
| 2 | Communication and Collaboration | Listening, audience analysis, technical writing, oral presentation, teamwork, feedback and conflict-aware communication. | Approx. 16 hours | CO2 | Module 2 |
| 3 | Ethics and Professional Responsibility | Values, duties, consequences, rights, safety, honesty, conflicts of interest, privacy and accountability. | Approx. 15 hours | CO3 | Module 3 |
| 4 | Sustainability, Policy and Public Impact | Environment-society relationships, sustainable development, public participation, policy context and impact assessment. | Approx. 15 hours | CO4 | Module 4 |

LEARNING ROADMAP

# How the Modules Connect

### Humanities, Society and Engineering

Human experience, culture, society, institutions, technology, engineering context and critical inquiry.

CO1 · Approx. 14 hours

### Communication and Collaboration

Listening, audience analysis, technical writing, oral presentation, teamwork, feedback and conflict-aware communication.

CO2 · Approx. 16 hours

### Ethics and Professional Responsibility

Values, duties, consequences, rights, safety, honesty, conflicts of interest, privacy and accountability.

CO3 · Approx. 15 hours

### Sustainability, Policy and Public Impact

Environment-society relationships, sustainable development, public participation, policy context and impact assessment.

CO4 · Approx. 15 hours

MODULE 1

## Humanities, Society and Engineering

Human experience, culture, society, institutions, technology, engineering context and critical inquiry.

**Approx. 14 hours** **CO1** **Understand · Apply**

### Learning goals

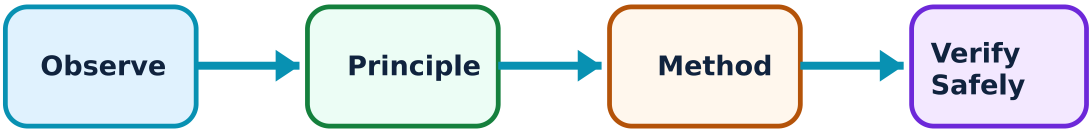
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Humanities, Society and Engineering: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Humanities and social-science perspectives−

Humanities explore meaning, culture, values, history and communication; social sciences investigate people, institutions, behaviour and collective systems. Together they help engineers understand the context in which technology is designed and used.

### Study and application method

For a technical proposal, identify affected people, institutional setting, cultural context and questions that cannot be answered by technical performance alone.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** For a technical proposal, identify affected people, institutional setting, cultural context and questions that cannot be answered by technical performance alone.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Do not assume a technically efficient solution is automatically acceptable, equitable or useful to users.

## 2. Technology and society−

Technologies reshape work, access, communication, environment and power relations. Social outcomes depend on design choices, deployment, governance and unequal resources—not on technology alone.

### Study and application method

Map stakeholders, benefits, burdens and likely unintended effects; distinguish direct users from indirect communities.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Map stakeholders, benefits, burdens and likely unintended effects; distinguish direct users from indirect communities.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Avoid technological determinism: people, policies and institutions influence outcomes.

## 3. Evidence and critical reasoning−

Responsible judgement uses credible evidence, transparent assumptions and reasoned interpretation. Claims need appropriate sources and must distinguish fact, inference and opinion.

### Study and application method

State claim, evidence, source quality and uncertainty; seek counterexamples before drawing a conclusion.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** State claim, evidence, source quality and uncertainty; seek counterexamples before drawing a conclusion.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Do not present a single anecdote or unverified online claim as general evidence.

**Module 1 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

## Communication and Collaboration

Listening, audience analysis, technical writing, oral presentation, teamwork, feedback and conflict-aware communication.

Approx. 16 hours | CO2 | Understand · Apply

### Learning goals

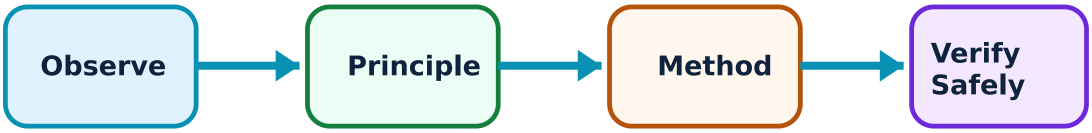
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Communication and Collaboration: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Audience-centred communication−

Effective communication adapts purpose, language, structure and evidence to the audience. A technician, user, manager and community member may need different levels of detail.

### Study and application method

Define audience, purpose and action requested; use a logical heading structure, plain language and visual or quantitative support where appropriate.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Define audience, purpose and action requested; use a logical heading structure, plain language and visual or quantitative support where appropriate.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Do not equate jargon with professionalism; unexplained technical language can exclude key stakeholders.

## 2. Technical writing and presentation−

Technical documents and presentations make claims traceable through clear structure, figures, data and references. Good delivery combines concise narrative with evidence and response to questions.

### Study and application method

Use problem → method → evidence → conclusion → limitation/recommendation sequence; rehearse timing and verify all figures/units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Use problem → method → evidence → conclusion → limitation/recommendation sequence; rehearse timing and verify all figures/units.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Never manipulate scales, omit key limitations or use others’ text/data without attribution.

## 3. Teamwork and feedback−

Teams coordinate knowledge, tasks and decisions. Constructive feedback focuses on work, evidence and improvement rather than personal judgement.

### Study and application method

Agree roles, norms and review points; record decisions and give feedback using observation → impact → actionable suggestion.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Agree roles, norms and review points; record decisions and give feedback using observation → impact → actionable suggestion.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Do not allow invisible labour or unaddressed conflict to undermine team responsibility.

**Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

# Ethics and Professional Responsibility

Values, duties, consequences, rights, safety, honesty, conflicts of interest, privacy and accountability.

**Approx. 15 hours** **CO3** **Understand · Apply**

### Learning goals

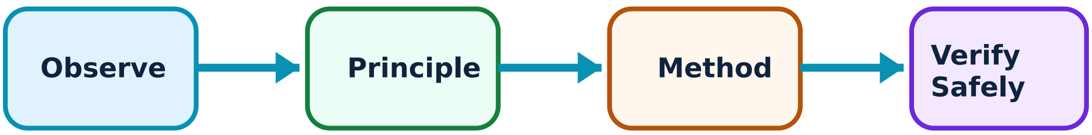
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Ethics and Professional Responsibility: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Ethical reasoning frameworks−

Ethical analysis can examine consequences, duties/rights, fairness, care and character. Structured comparison exposes assumptions and affected parties.

## Study and application method

Define dilemma, stakeholders and facts; analyse foreseeable harms/benefits, duties/rights and fairness; state a reasoned recommendation with uncertainty.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

## Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Define dilemma, stakeholders and facts; analyse foreseeable harms/benefits, duties/rights and fairness; state a reasoned recommendation with uncertainty.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Do not reduce ethics to personal preference or compliance alone; legal permission may still leave ethical concerns.

## 2. Safety, honesty and accountability−

Engineering responsibility includes truthful reporting, competence, safety margins, disclosure of uncertainty and willingness to escalate risk. Accountability requires documentation and clear decision ownership.

### Study and application method

Identify safety-critical assumptions, verification step, responsible role and escalation path in a technical decision.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Identify safety-critical assumptions, verification step, responsible role and escalation path in a technical decision.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Never conceal test failures, altered data or conflicts of interest to meet a deadline.

## 3. Privacy, inclusion and fairness−

Technology can expose personal data, exclude users or distribute risk unevenly. Inclusive design and responsible data practices consider diverse abilities, access and potential discrimination.

### Study and application method

Perform an impact check: whose data, access, language, ability, cost or risk could be affected, and what mitigation is possible?

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Perform an impact check: whose data, access, language, ability, cost or risk could be affected, and what mitigation is possible?

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Do not collect, share or retain personal information without legitimate purpose and adequate protection.

**Module 3 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

## Sustainability, Policy and Public Impact

Environment-society relationships, sustainable development, public participation, policy context and impact assessment.

Approx. 15 hours | CO4 | Understand · Apply

### Learning goals

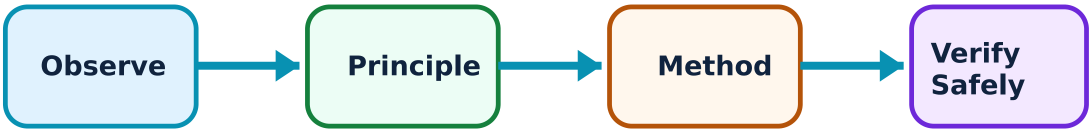
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Sustainability, Policy and Public Impact: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Environment and society−

Environmental conditions shape health, livelihoods and social equity, while human activities alter air, water, land and climate. Engineering choices create both direct and indirect environmental impacts.

### Study and application method

Map a system life cycle from materials to end-of-life, identify affected communities and state at least one measurable environmental indicator.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Map a system life cycle from materials to end-of-life, identify affected communities and state at least one measurable environmental indicator.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**Common mistake / precaution:** Do not move an environmental burden from one place or group to another and call the result sustainable.

## 2. Sustainable development−

Sustainable development considers present needs without undermining future ability to meet needs. It requires balancing technical, economic, environmental and social dimensions transparently.

### Study and application method

Compare design alternatives against stated criteria such as safety, lifetime cost, energy, materials, maintenance, accessibility and emissions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Compare design alternatives against stated criteria such as safety, lifetime cost, energy, materials, maintenance, accessibility and emissions.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** [illegible] concept [illegible] [illegible] [illegible]. [illegible] diagram / circuit / formula / procedure [illegible] [illegible]. [illegible] result [illegible] application [illegible] [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Common mistake / precaution:** Do not label a proposal sustainable without defining criteria, evidence and trade-offs.

## 3. Public impact and policy−

Policies, standards, regulations and public participation influence which technical options are feasible and legitimate. Engineers communicate evidence but should recognise the roles of communities and institutions in decisions.

### Study and application method

Identify relevant stakeholder groups, decision authority, evidence needed and a respectful participation or feedback method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Identify relevant stakeholder groups, decision authority, evidence needed and a respectful participation or feedback method.

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**Malayalam support:** □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**Common mistake / precaution:** Do not treat public engagement as a formality after key decisions are irreversible.

**Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

## Rapid Recall Bank

### Recall 1

State symbols and SI units before substitution.

### Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

### Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

### Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

### Recall 5

For a comparison: use construction, working, result, advantage and limitation.

### Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

## Diagram Library for Rapid Revision

### Module 1: Humanities, Society and Engineering

Use the concept flow to revise observation, principle, method and verification.

### Module 2: Communication and Collaboration

Use the concept flow to revise observation, principle, method and verification.

### Module 3: Ethics and Professional Responsibility

Use the concept flow to revise observation, principle, method and verification.

### Module 4: Sustainability, Policy and Public Impact

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

## Official Self-Learning and Student Activities

### Structured activity

Analyse a local technology or infrastructure issue using a stakeholder-impact map.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Write a one-page technical brief for a non-specialist audience with cited evidence.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Discuss an engineering-ethics case using consequences, duties and fairness.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Compare two design options using environmental, social, technical and cost criteria.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

**Activity discipline:** The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

# Assessment Methodology and Examination Pattern

| Official assessment structure | | |
|---|---|---|
| **Component** | **Official mark / conversion** | **Student evidence** |
| Attendance | 5 marks | End-of-semester attendance component. |
| Self-learning activities | 15 marks | Module-linked activities. |
| Series examinations | 20 + 20 converted | Two 50-mark series examinations. |
| CIE total | Theory CIE 40 | Continuous internal evaluation. |
| Written ESE | Theory ESE 60 | 2.5 hours; official Part A / Part B / Part C pattern. |

### Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

### Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

# Module-wise Questions with Answers

## Module 1 — Humanities, Society and Engineering

**Explain Humanities and social-science perspectives and state one correct method, application or precaution.−**

Humanities explore meaning, culture, values, history and communication; social sciences investigate people, institutions, behaviour and collective systems. Together they help engineers understand the context in which technology is designed and used.

**Answer framework:** For a technical proposal, identify affected people, institutional setting, cultural context and questions that cannot be answered by technical performance alone.

**Important point:** Do not assume a technically efficient solution is automatically acceptable, equitable or useful to users.

**Explain Technology and society and state one correct method, application or precaution.−**

Technologies reshape work, access, communication, environment and power relations. Social outcomes depend on design choices, deployment, governance and unequal resources—not on technology alone.

**Answer framework:** Map stakeholders, benefits, burdens and likely unintended effects; distinguish direct users from indirect communities.

**Important point:** Avoid technological determinism: people, policies and institutions influence outcomes.

**Explain Evidence and critical reasoning and state one correct method, application or precaution.−**

Responsible judgement uses credible evidence, transparent assumptions and reasoned interpretation. Claims need appropriate sources and must distinguish fact, inference and opinion.

**Answer framework:** State claim, evidence, source quality and uncertainty; seek counterexamples before drawing a conclusion.

**Important point:** Do not present a single anecdote or unverified online claim as general evidence.

## Module 2 — Communication and Collaboration

**Explain Audience-centred communication and state one correct method, application or precaution.−**

Effective communication adapts purpose, language, structure and evidence to the audience. A technician, user, manager and community member may need different levels of detail.

**Answer framework:** Define audience, purpose and action requested; use a logical heading structure, plain language and visual or quantitative support where appropriate.

**Important point:** Do not equate jargon with professionalism; unexplained technical language can exclude key stakeholders.

**Explain Technical writing and presentation and state one correct method, application or precaution.−**

Technical documents and presentations make claims traceable through clear structure, figures, data and references. Good delivery combines concise narrative with evidence and response to questions.

**Answer framework:** Use problem → method → evidence → conclusion → limitation/recommendation sequence; rehearse timing and verify all figures/units.

**Important point:** Never manipulate scales, omit key limitations or use others' text/data without attribution.

**Explain Teamwork and feedback and state one correct method, application or precaution.−**

Teams coordinate knowledge, tasks and decisions. Constructive feedback focuses on work, evidence and improvement rather than personal judgement.

**Answer framework:** Agree roles, norms and review points; record decisions and give feedback using observation → impact → actionable suggestion.

**Important point:** Do not allow invisible labour or unaddressed conflict to undermine team responsibility.

## Module 3 — Ethics and Professional Responsibility

**Explain Ethical reasoning frameworks and state one correct method, application or precaution.−**

Ethical analysis can examine consequences, duties/rights, fairness, care and character. Structured comparison exposes assumptions and affected parties.

**Answer framework:** Define dilemma, stakeholders and facts; analyse foreseeable harms/benefits, duties/rights and fairness; state a reasoned recommendation with uncertainty.

**Important point:** Do not reduce ethics to personal preference or compliance alone; legal permission may still leave ethical concerns.

**Explain Safety, honesty and accountability and state one correct method, application or precaution.−**

Engineering responsibility includes truthful reporting, competence, safety margins, disclosure of uncertainty and willingness to escalate risk. Accountability requires documentation and clear decision ownership.

**Answer framework:** Identify safety-critical assumptions, verification step, responsible role and escalation path in a technical decision.

**Important point:** Never conceal test failures, altered data or conflicts of interest to meet a deadline.

**Explain Privacy, inclusion and fairness and state one correct method, application or precaution.−**

Technology can expose personal data, exclude users or distribute risk unevenly. Inclusive design and responsible data practices consider diverse abilities, access and potential discrimination.

**Answer framework:** Perform an impact check: whose data, access, language, ability, cost or risk could be affected, and what mitigation is possible?

**Important point:** Do not collect, share or retain personal information without legitimate purpose and adequate protection.

## Module 4 — Sustainability, Policy and Public Impact

**Explain Environment and society and state one correct method, application or precaution.−**

Environmental conditions shape health, livelihoods and social equity, while human activities alter air, water, land and climate. Engineering choices create both direct and indirect environmental impacts.

**Answer framework:** Map a system life cycle from materials to end-of-life, identify affected communities and state at least one measurable environmental indicator.

**Important point:** Do not move an environmental burden from one place or group to another and call the result sustainable.

**Explain Sustainable development and state one correct method, application or precaution.−**

Sustainable development considers present needs without undermining future ability to meet needs. It requires balancing technical, economic, environmental and social dimensions transparently.

**Answer framework:** Compare design alternatives against stated criteria such as safety, lifetime cost, energy, materials, maintenance, accessibility and emissions.

**Important point:** Do not label a proposal sustainable without defining criteria, evidence and trade-offs.

**Explain Public impact and policy and state one correct method, application or precaution.−**

Policies, standards, regulations and public participation influence which technical options are feasible and legitimate. Engineers communicate evidence but should recognise the roles of communities and institutions in decisions.

**Answer framework:** Identify relevant stakeholder groups, decision authority, evidence needed and a respectful participation or feedback method.

**Important point:** Do not treat public engagement as a formality after key decisions are irreversible.

PRACTICE ONLY

# Practice Model Paper

**Important:** This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

## Part A — short response

1. State one key definition from Module 1: Humanities, Society and Engineering.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Communication and Collaboration.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Ethics and Professional Responsibility.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Sustainability, Policy and Public Impact.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

### Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

### Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Human experience, culture, society, institutions, technology, engineering context and critical inquiry..
2. Develop a complete answer or practical plan for: Listening, audience analysis, technical writing, oral presentation, teamwork, feedback and conflict-aware communication..
3. Develop a complete answer or practical plan for: Values, duties, consequences, rights, safety, honesty, conflicts of interest, privacy and accountability..
4. Develop a complete answer or practical plan for: Environment-society relationships, sustainable development, public participation, policy context and impact assessment..

LAST-MILE REVISION

## Quick Revision

### Module 1: Humanities, Society and Engineering

Human experience, culture, society, institutions, technology, engineering context and critical inquiry.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 2: Communication and Collaboration

Listening, audience analysis, technical writing, oral presentation, teamwork, feedback and conflict-aware communication.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 3: Ethics and Professional Responsibility

Values, duties, consequences, rights, safety, honesty, conflicts of interest, privacy and accountability.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 4: Sustainability, Policy and Public Impact

Environment-society relationships, sustainable development, public participation, policy context and impact assessment.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

**Common mistakes:** Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

**Exam checklist:** Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

## Glossary

| Important terms from the syllabus | |
|---|---|
| **Term / topic** | **One-line meaning** |
| Humanities and social-science perspectives | Humanities explore meaning, culture, values, history and communication; social sciences investigate people, institutions, behaviour and collective systems. |
| Technology and society | Technologies reshape work, access, communication, environment and power relations. |
| Evidence and critical reasoning | Responsible judgement uses credible evidence, transparent assumptions and reasoned interpretation. |
| Audience-centred communication | Effective communication adapts purpose, language, structure and evidence to the audience. |
| Technical writing and presentation | Technical documents and presentations make claims traceable through clear structure, figures, data and references. |
| Teamwork and feedback | Teams coordinate knowledge, tasks and decisions. |
| Ethical reasoning frameworks | Ethical analysis can examine consequences, duties/rights, fairness, care and character. |
| Safety, honesty and accountability | Engineering responsibility includes truthful reporting, competence, safety margins, disclosure of uncertainty and willingness to escalate risk. |
| Privacy, inclusion and fairness | Technology can expose personal data, exclude users or distribute risk unevenly. |
| Environment and society | Environmental conditions shape health, livelihoods and social equity, while human activities alter air, water, land and climate. |
| Sustainable development | Sustainable development considers present needs without undermining future ability to meet needs. |
| Public impact and policy | Policies, standards, regulations and public participation influence which technical options are feasible and legitimate. |

LEARNING RESOURCES

# References

### Recommended humanities and professional-ethics references


1. Charles E. Harris Jr., Michael S. Pritchard and Michael J. Rabins, Engineering Ethics.
2. Mike W. Martin and Roland Schinzinger, Ethics in Engineering.
3. UNESCO, Engineering for Sustainable Development guidance.


### Approved public humanities and social-science sources

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